

The Conservation of Nature in Viet Nam

by

EGBERT W. PFEIFFER, M.A.(UBC), Ph.D.(California at Berkeley)

Professor of Zoology, University of Montana, Missoula, Montana 59812, USA.

INTRODUCTION

During more than thirty years of warfare, which latterly has included deliberate destruction of many of its fields and forests, Viet Nam was nevertheless initiating measures to preserve its natural resources. Actions for conservation of Nature were begun in the Democratic Republic of Viet Nam (DRV, the so-called North Viet Nam) as early as 1960, when their scientists 'discovered' an area of undisturbed primeval forest 120 km south-west of Hanoi. By 1962, the government of the DRV had decreed that the 25,000-hectares' area be a National Park, the first one in all of Indochina.

I first had contact with this Park in the summer of 1973, when Professor Arthur H. Westing and I revisited Viet Nam to continue our studies of the war's effects on Indochina's environment (cf. Pfeiffer, 1970; Westing & Westing, 1981). We were invited to spend a day inspecting the Park, which turned out to be a spectacular example of undisturbed climax (triple-canopy) rain-forest.

In January of 1983, a second opportunity arose for me to visit the Park for a longer period. In addition to learning more about this National Park, named Cuc Phuong, I hoped to learn about any activities or plans that the new government of Viet Nam (Socialist Republic of Viet Nam) had for conservation of Nature throughout the country. Were new parks planned? What about endangered species in Viet Nam? Was hunting regulated? What was the effect of the slash-and-burn, semi-nomadic culture of the mountain-dwelling, ethnic minority people on the flora and fauna?

CURRENT FOREST PROBLEMS

The overall problems facing Viet Nam's Ministry of Forestry, as explained to me, are related to Viet Nam's changing economy; that is, the change from war to peace. Specifically, the question is how to relate forest management to the economy as a whole. 'Forests for the people' is the slogan which, however, requires much planning to put into practice. In the South, vast areas of coastal mangrove forest and inland hardwood forests were either totally destroyed or severely degraded by US chemical warfare and high-explosives. There are decisions to be made as to whether to replant these forests, and, if so, what species to replant, or alternatively to open the devastated, former forested areas for crop production. Already work is under way to restore the productivity of the mangrove forests. Dr Bui Thi

Lang, a marine biologist (graduate of the University of California at La Jolla), is now engaged in a successful effort to develop the culture of shrimps for export.

THE INSTITUTE OF FOREST INVENTORY AND PLANNING (IFIP)

It is the job of the IFIP to solve the above and many other problems. In discussions with officials, I learned that the Institute was responsible: for making investigations on forests, forest soils, and forest resources (wood, timber, birds, and other wildlife); for making forest inventories and assessing the changes, the growth and damages, that occurred in these fields; for carrying out classifications of forest regions and planning the development of the forestry sector for the whole country, including the planning of the development of forestry in the forest regions and provinces, and in the economic zones of forestry and the national parks; for making forest inventories and drawing up the working plans for the management of the forest in forest enterprises; for carrying out research in the fields of forest inventory, forest replanting, and working plans; and for introducing new technical advances, so as to increase the quality and efficiency of the forest-inventory planning work.

As a zoologist, I was particularly interested in the activities related to wildlife. The mandate of the organization is 'To make investigations on birds and wildlife resources of the country, and in various zones, and national parks of nature conservancy, to study the ecology of birds and wildlife of economical importance.'

To do all this, IFIP employs about 5,000 people, including scientists who had trained at the University of Hanoi, or overseas in the Netherlands or India. Mr Do Tuoc, a zoologist, was my host at the Institute headquarters during our day-long stay. His special interest was the bird life of Viet Nam, and he showed me some of the 300 species he had collected so far out of the total of *ca* 1,000 species that are known to live in Viet Nam. He was working for the establishment of bird sanctuaries in Viet Nam.

REGULATION OF HUNTING

The word 'sanctuaries' brought to mind the problem of hunting in Viet Nam. As both a hunter and government official, Mr Tuoc was eager to answer my questions. Before 1963, there were no laws regarding hunting. Since then, there has been, starting in the north, a Department of Protection within the Ministry of

Forestry. It has wardens who attempt (not too successfully, said Tuoc) to enforce the laws. Hunters are supposed to have a license and can legally hunt only in December and January, when reproductive activity of most hunted species is low. Popular species for hunting are Wild Deer (*Cervus unicolor*) and Wild Pigs (*Sus scrofa*). Tigers (*Panthera tigris*) and Leopards (*P. pardus*) can be hunted the year around. But twenty-eight species of birds and mammals may not be hunted at all. These include the Ring-neck Pheasant (*Phasianus colchicus*) and several very interesting mammals—such as the Indochina Gibbon (*Hylobates concolor*) and the Douc Langur (*Pygathrix nemaeus*), 'the world's most colourful primate', which is found in the triple-canopy forests of Indochina.

The list of the banned species also includes the wild cattle of Indochina: the Kouprey (*Bos sauveli*), Banteng (*Bos banteng*), and the Gaur (*Bos gaurus*). The Vietnamese zoologist Professor Vo Quy informed us that possibly 10 individual Kouprey still remain alive in Viet Nam, and the prospects for its survival are indeed not good. A loss of this wild cow will be a tragedy because it is thought to be the species from which many of our domestic cattle are derived. Hunting of the Javan Rhinoceros (*Rhinoceros sondaicus*) also is banned, and the Vietnamese estimate there may be only five individuals of this species left in Viet Nam.

The Asian Elephant (*Elephas maximus*) is also protected, although there are certain exceptions. For example, in the central highlands of Viet Nam, it has long been a practice to capture Elephants and tame them. They are used in tracking, hunting, and capturing, other Elephants. Before the Second Indochina war, during the French occupation, the Central Highlands area was the biggest elephant market in the whole of South-east Asia, supplying log-hauling Elephants to Thailand, Burma, Laos, and Cambodia, and circus Elephants and zoo Elephants to Europe. During the Indochina wars, this market fell off and Elephants were mainly used for transporting supplies to the revolutionary army. Expectedly, many Elephants were therefore hunted down and killed during the war in Viet Nam. However, today in the Highlands, the mountain people, with the help of the Government, are making concerted efforts to increase the numbers of Elephants, both for commercial purposes and for increasing the wild herd (Hung, 1982).

Mr Tuoc also pointed out that, because of the many problems brought on by the war, the Government of Viet Nam is preparing new laws on hunting. For example, Tiger habitat has been reduced as a result of land-clearing which occurred during the war in order to destroy cover for the guerillas, and as a consequence of increased human population. Many people now hunt Tigers, and in some areas the Government tries to limit the Tiger kill to four in all per year. There is apparently a big market for Tiger bone-marrow for 'medicinal purposes'.

Many species of birds are captured by means of traps, nets, glues, and, in the mountains, by crossbow. As yet there are no special regulations regarding methods of capture of birds which are used for food, medicine, and house-pets, according to zoologist Tuoc.

Mr Tuoc had been trained in the Department of Zoology, University of Hanoi, by Professor Dao Van Tien and his colleague Dr Vo Quy, who showed us the zoological museum, laboratories, and classrooms (Fig. 1). My impression of their facilities are similar to those of Dr John D. Constable, of Harvard University, who recently wrote about his visit to Professor Tien's establishment as follows: 'It is reminiscent of that of Cuvier or Darwin where we were surrounded by books, reprints, skeletons, horns, and piles of great tin boxes containing skins, with here and there a few live animals, including a tortoise and a squirrel' (Constable, 1982).

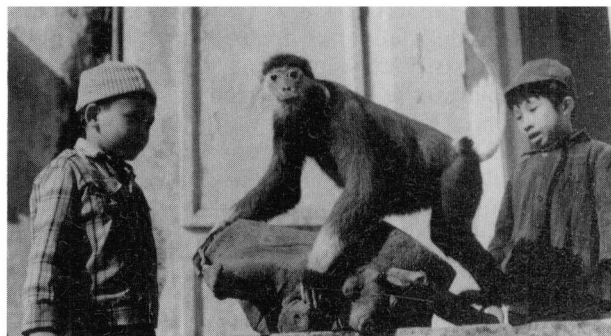


FIG. 1. Preserved Douc Langur; from the collection of mammals of the Department of Zoology of the University of Hanoi. Photo: E.W. Pfeiffer.

CUC PHUONG NATIONAL PARK

Following our stay in Hanoi, visiting IFIP, the University, and the new zoological park (which had a good collection of Vietnamese fauna housed in spacious, new cages), my colleagues and I began our trip to Cuc Phuong National Park, 120 km south-west of Hanoi. We travelled across part of the remarkably fertile delta of the Red River, and then began a slow ascent into the foothills of the northern Annamite Mountains. These mountains are mostly of limestone of the Triassic period. After winding up an exceedingly rough, one-lane track, we went through a narrow pass and then out onto a broad flat valley surrounded by towering limestone hills and mountains. All but the steepest cliffs were covered with a rich green canopy of dense vegetation. After meeting with officials of the Park, our party was briefed about its main physical characteristics and other aspects. It is an area containing two ranges of mountains, separated by a valley which is wide at its mouth but becomes a very narrow canyon as it proceeds eastwards. It is apparently this very narrow canyon which has prevented the mountain people from entering much of the forest, to destroy it in their slash-and-burn culture. That is why Cuc Phuong still has its unique, largely virgin flora and vegetation.

Physical and Biological Data

The altitude at the entrance of the Park is about 100 m, and the tallest peak is 630 m high. The average temperature in the winter is 9°C, with the coldest

recorded reading at 0.18°C, so that it seems probable that frost occurs in the higher parts of these mountains. The average annual temperature is 21°C, and the highest recorded temperature is 35°C. There is very heavy rainfall: some 2.2 m per year, although there are two dry months, December and January, in all with less than 100 mm of rainfall. During the rainy season the water from the mountains is absorbed by the soil, so that no river drains the valley of Cuc Phuong National Park.

The Vietnamese botanists in the Park have recorded about 1,700 species of vascular plants as growing in the Park—including seven hitherto undescribed species that appear to be endemic to the Park—and there are still about 200 different species that have not yet been identified. The biggest tree in the Park has a diameter of some 4 m at the base. Park zoologists have identified 137 species of birds, 65 species of mammals, 37 species of reptiles, and 16 species of amphibians, as living in its area. They have also found a new species of fish, living in pools of the vast caves that have become dissolved out of the limestone mountains. Among the indigenous mammals and birds found in the Park are Asiatic Black Bears (*Ursus thibetanus*), Tigers, Leopards, Wild Pigs, and two new species of bat, including one new genus; also Silver Pheasant (*Gennaesus nycthemerus*), and what the Vietnamese call a cave bird, which is a very large hornbill.

Muong Minority Inhabitants

After some instruction in the Park's natural history, we drove up the centre of the valley on a dirt road that took us past many thatched houses on stilts (Fig. 2). These were distinct from the Vietnamese houses, the floors of which are on the ground. The stilted houses were those of the ethnic minority Muong people, who were among the earliest human inhabitants of the area.

They live by an age-old 'slash-and-burn' culture, which consists of small family groups clearing a very few hectares of forest, burning the downed timber, and planting it to mountain rice, which needs no irrigation. After four or five years, the soil is exhausted and they move on, to repeat the cycle elsewhere. There are some 1,000,000 of the mountain people, in about 100 different tribes*, speaking different dialects, inhabiting the mountains of all Viet Nam. Within the Park, there are some 1,000 men, women, and children, of the Muong tribe. Therein lies the difficulty for management of the Park, because these people are still practising slash-and-burn culture. Although much of the Park is inaccessible to them, there are considerable areas which will become seriously degraded if this practice continues much longer.

Furthermore, these Muong people carry out extensive 'illegal' hunting. It was exciting to meet a young Muong hunter with his cross-bow and quiver of bamboo arrows as we continued to traverse the forest (Fig. 3). He said that he was hunting birds, which is illegal in the Cuc Phuong. Apparently the Vietnamese authorities in the Park overlook these transgressions because there is a unique relationship between these people and the other Vietnamese—and after all, the Muong were the original inhabitants of this land. The authorities will soon make a decision as to whether to move these *ca* 1,000 people out of the Park into nearby adjacent foothills, where they would try to establish them in a stable agricultural system, or to convince them to establish such a system within a limited area in the Park.

*In answer to our query, Professor Pfeiffer replied (*in litt.* 14 August 1984) 'The Muong are one of the [ca] 100 different ethnic minority groups (tribes)'.—Ed.



FIG. 2. Muong house in Cuc Phuong National Park. Photo: E.W. Pfeiffer.



FIG. 3. *Muong cross-bow hunter in Cuc Phuong National Park.*
Photo: E.W. Pfeiffer.

In addition to hunting and growing mountain rice, these people raise many chickens, pigs, and cattle. They also have problems with Tigers in the Park killing the Muong domestic stock. These people often tip their bamboo arrows with a toxic substance from a local species of tree, reminiscent in practice of the curare used by South American natives. Another problem facing the administrators of the Park is the increasing population of the Muong minority, because the women now go to Government hospitals to have their babies, which results in an increased effective birth-rate. It is the policy of the Vietnamese Government strongly to recommend a variety of birth-control methods; however, the programme apparently has little effect among the Muong.

The Big Cave

After visiting briefly with the Muong community, we proceeded to a well-worn trail leading into the forest and up to the famous Cuc Phuong Park Cave. We had to climb some 600 rungs of ladder, built to facilitate tourists' visits to this awesome place. The Cave had been dissolved out of the limestone, and is a truly huge and complicated underground river-and-cave system. At the mouth it is from 3 to 4 m high and over 30 m across the floor (Fig. 4). There are ancient artworks on the wall of the dome and, at the time of discovery, skeletons of primitive humans were found in three ancient tombs, with Neolithic tools, dating back some 8,000 years—according to European anthropologists who were brought in by the Vietnamese to help them to learn about their ancient predecessors.

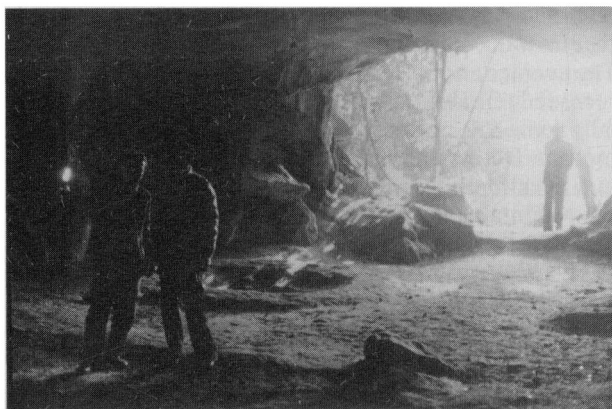


FIG. 3. *Muong cross-bow hunter in Cuc Phuong National Park.*
Photo: E.W. Pfeiffer.

Following the trip to the big Cave, we enjoyed a comfortable night in the local guest-house, which was built on stilts in the style of the Muong people, and could comfortably house a dozen guests. The Park has about 7,000 visitors per year, one-third of whom are foreigners—mostly scientists engaged in a wide variety of biological studies.

PARK RESEARCH

The Park's research facilities include the zoological museum and its many well-mounted specimens of the vertebrate and invertebrate life of the Park (Fig. 5). Plantations of trees (Fig. 6), grown for export to different parts of Viet Nam, and a deer-breeding station, are among the principal activities. The zoologist in charge of this deer-breeding programme stated that they were trying to develop a larger deer to release into the forest. Another purpose of the deer-farm was to collect antlers in the velvet. After 45 days of growth, the antlers are cut off and dissolved in a traditional way* in order to make a preparation known as Pantocrin, which is exported to various countries in Asia and Europe. It is reported to have medicinal properties, that are especially beneficial for old people.



FIG. 5. *Inside the museum at Cuc Phuong National Park.* Photo: E.W. Pfeiffer.

*In answer to our query about the 'traditional way', the Author replied that he 'cannot add to this—sorry'.—Ed.



FIG. 6. *Tree plantation in Cuc Phuong National Park. Photo: E.W. Pfeiffer.*

One can take heart in knowing that a unique and precious part of our globe is in good administrative hands—that maintenance and prospects for protection looked bright at the time of our visit. Problems remain, however. As one Vietnamese put it, 'Preserving the primeval forest of Cuc Phuong is a difficult, complex task due to the poor scientific and technical equipment in the country, and also to the local population's long practice of slash-and-burn' (Fig. 7). 'That is why', he said, 'international assistance is very necessary.' It is ardently to be hoped that international organizations such as the World Wildlife Fund will come to the assistance of these very dedicated, but also very poor, people, who are so anxious to rebuild on the remains of their ravaged land.

ACKNOWLEDGEMENTS

During my January 1983 visit to Viet Nam, as in all past visits, the authorities were most helpful in arranging trips to institutions involved in the conservation of Nature. I thank officials of the Department of Zoology in the University of Hanoi, of the Ministry of Forestry Institute of Forest Inventory and Planning (IFIP), of



FIG. 7. *Forest being cleared for 'slash-and-burn' cultivation by people of the Muong tribe in Cuc Phuong National Park. Photo: E.W. Pfeiffer.*

the new zoological park in Hanoi, and, finally, of Cuc Phuong National Park, for their hospitality and help.

SUMMARY

Systematic efforts to conserve Viet Nam's natural resources began (in the North) in the early 1960s with the establishment of a 25,000-hectares' area, Cuc Phuong National Park, south-west of Hanoi. Since the end of the Second Indochina war, Viet Nam's Ministry of Forestry has undertaken a comprehensive programme to restore forests that were destroyed or damaged in the fighting, to inventory forests and their resources, and to develop forest management for the economy and recreation of the Vietnamese people. The Ministry of Forestry's Institute of Forest Inventory and Planning is the agency carrying out these tasks; it employs about 5,000 people, including well-trained scientists.

Before 1963, there were no regulations regarding hunting in Viet Nam. Now the Ministry of Forestry has a Department of Protection which attempts to regulate hunting of game, such as deer, Wild Pigs, Tigers, and Leopards. Many species of birds and mammals are now totally protected. These include two primates, the Indochina Gibbon and the Douc Langur, as well as three species of wild cattle, the Kouprey, Banteng, and Gaur. The Kouprey and the Javan Rhinoceros are classified as endangered species because only a few individuals are known to be still alive, at least in Viet Nam. Elephants are trapped and trained for commercial purposes and for capture of wild Elephants, while the wild herds are increasing owing to the war's end and the prevention of killing.

Cuc Phuong National Park lies 120 km south-west of Hanoi, in the Annamite Mountains. Much of the Park is virgin tropical rain-forest, and it contains in all about 1,700 species of vascular plants, some of which are apparently endemic to the area. There are 137 species of birds, 65 species of mammals, and many different species of reptiles and amphibians, in the Park. An excellent museum, tree plantations, and a wild-animal breeding programme, are part of the continuing research activities.

Some 1,000 Muong ethnic-minority people still live in the Park, creating problems because of their semi-nomadic 'slash-and-burn' culture. Park officials expressed their deep desire for cooperation with international conservation organizations such as the World Wildlife Fund.

REFERENCES

- CONSTABLE, J.D. (1982). Visit to Viet Nam. *Oryx*, **16**(3), pp. 249–54.
- HUNG, Vu (1982). Elephant hunters in the Central Highlands. *Viet Nam Courier*, **18**(6), pp. 27–31.
- PFEIFFER, E.W. (1970). Defoliation and bombing effects in Viet Nam. *Biological Conservation*, **2**(2), pp. 149–51, 2 figs.
- WESTING, A.H. & WESTING, C.E. (1981). Endangered species and habitats of Viet Nam. *Environmental Conservation*, **8**(1), pp. 59–62.